

SEPTEMBER 13, 2026

UNSUPERVISED LEARNING METHOD

ASSIGNMENT: ASSOCIATION RULE LEARNING

Dataset: Groceries

Software: R, Python, or any appropriate software

Objective

The purpose of this assignment is to help you understand how **association rule learning** can be used to discover relationships between products that customers purchase together.

Using the **Groceries dataset**, perform an association rule analysis and answer the following questions.

Questions

Question 1: Understand the Data

Explore the Groceries dataset and report:

- a. How many transactions are in the dataset?
- b. How many different grocery items are included?
- c. What are the **10 most frequently purchased items**?

Question 2: Frequent Itemsets

Identify some products that are frequently purchased together.

- a. Find at least **10 frequent itemsets** containing two or more products.
- b. Report their support values.
- c. Which itemset has the highest support?

d. What does the support value tell you about the customers' purchasing behavior?

Question 3: Association Rules

Generate association rules from the Groceries dataset.

a. Generate at least **10 association rules**.

b. For each rule, report:

- Antecedent
- Consequent
- Support
- Confidence
- Lift

Question 4: Understanding the Measures

Choose **three association rules** and explain:

a. What does the **support** of each rule mean?

b. What does the **confidence** of each rule mean?

c. What does the **lift** of each rule mean?

Question 5: Find Interesting Rules

From all the rules you generated:

a. Identify the rule with the **highest confidence**.

b. Identify the rule with the **highest lift**.

c. Identify the rule with the **highest support**.

d. Are these the same rule? Explain your answer.

Question 6: Interpret a Rule

Select one interesting rule, for example:

$\{\text{Product A, Product B}\} \rightarrow \{\text{Product C}\}$

Explain the rule in simple language.

For example:

Customers who purchase Product A and Product B also tend to purchase Product C.

Then explain whether the rule represents a **strong or weak association** and why.

Question 7: Business Application

Imagine that you are a supermarket manager.

Based on your results:

- Identify **two products that could be placed close to each other**.
- Identify **one possible product bundle**.
- Give one example of how the supermarket could use association rules to recommend products to customers.

Question 8: Understanding Association

Answer the following:

- Does an association rule prove that one product **causes** customers to buy another product? Explain.
- Why might a rule have high confidence but low lift?
- What does a lift value **greater than 1** indicate?

Submission

Submit:

1. A short report containing your answers.
2. The code used for your analysis.
3. Tables or figures showing your important results.